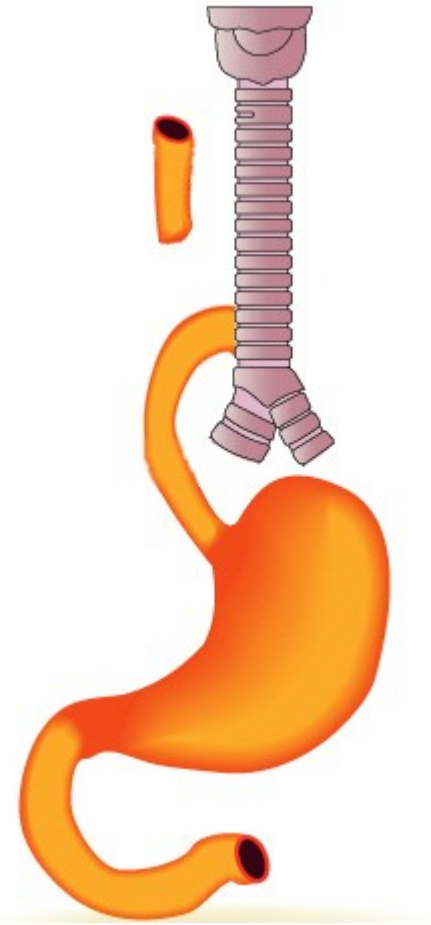
Most infants with a tracheoesophageal fistula (TEF) have esophageal atresia as pictured in the image on the right. The disorder is usually discovered on the first day of life when an NG tube cannot be passed beyond 8-10 cm, but air is evident within the stomach on x-ray. The disorder presents a situation where gastric contents are easily aspirated into the trachea and air enters into and distends the stomach, resulting in compromised respiration.

Corrective surgery for a TEF usually involves placing the infant in a left-lateral decubitus position. A 4th interspace thoracotomy is performed. In order to maintain adequate oxygenation, the surgery may have to be performed in 3-5 minutes bursts with 1-2 minutes of time in between to allow for ventilation without retractors in place. An NG tube is usually placed by the anesthetist after anastomosis of the posterior esophageal wall.

Patients presenting for this procedure often have significant cardiac abnormalities that must be taken into account. If the patient is premature, a high FiO2 that could predispose the patient to retinopathy should be avoided, if possible.



86% of congenital tracheoesophageal fistulas exhibit esophageal atresia with a distal fistula.

During repair of a tracheoesophageal fistula, the endotracheal tube should be advanced intentionally into the right mainstem bronchus and then slowly withdrawn until the first point at which breath sounds are heard in both lung fields. Then the bevel of the tube should be rotated so that it faces anteriorly. This is done to prevent ventilation of the fistula. Placement of the precordial stethoscope over the left lung alerts the anesthetist in the event the endotracheal tube migrates into the right mainstem bronchus. Tracheoesophageal fistula is associated with subglottic stenosis which may require placement of a smaller than usual endotracheal tube.

Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 1287-1291.

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